

Kushar Dogra

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SUMMARY

Cybersecurity graduate student with 1 year of web development experience and hands-on security training in penetration testing, binary exploitation, IDS/IPS, and incident response. Implemented security controls across 4 production applications serving 300+ users with zero critical vulnerabilities. Proficient with Burp Suite, Metasploit, Nessus, Splunk, Wireshark, Python.

WORK EXPERIENCE

Full Stack Developer

Aug 2023 – Aug 2024

Quindl Technologies, Coimbatore, India

- Delivered 4 production web applications (e-commerce, booking system, admin dashboards) to 300+ users by integrating security controls and conducting risk assessments, completing all projects within 4-month deadline with zero critical vulnerabilities in post-deployment scans
- Architected and secured 4 database schemas implementing role-based access controls (RBAC) with strict admin/user privilege separation, preventing unauthorized access across 300+ user accounts with zero privilege escalation incidents over 12-month period
- Led security-focused code reviews across 5-10 developer team, identifying and remediating 25+ security misconfigurations (SQL injection, XSS, auth bypasses) and reducing code-level vulnerabilities by 40% through stakeholder training and secure coding enforcement
- Implemented defense-in-depth security architecture using Django framework (CSRF tokens, XSS filtering, input sanitization, secure sessions, HTTPS enforcement), validating protections through automated vulnerability scans and achieving 95% OWASP Top 10 compliance across 4 production applications

CYBERSECURITY PROJECTS

Binary Exploitation & Vulnerability Research

- Developed exploit code for 6+ buffer overflow challenges targeting SUID binaries through custom shellcode injection, ret2libc attacks, and stack canary bypasses, achieving 96.5% CTF score and demonstrating advanced memory corruption expertise

Network Security & Penetration Testing

- Conducted end-to-end penetration tests across 5+ multi-VM lab environments following industry methodology (reconnaissance → exploitation → post-exploitation), compromising 100% of target systems through Metasploit framework and achieving full flag capture within time constraints
- Performed OSINT reconnaissance on 10+ target domains using Spiderfoot, Shodan, and Censys to map attack surface, then executed Hydra-based brute-force attacks achieving 80% success rate against weak SSH/web credentials in lab environments
- Architected 3 enterprise-grade firewall environments (IPFire, iptables) implementing NAT, stateful traffic filtering, rate limiting, and packet logging; validated wireless security by capturing and decrypting WPA2 handshakes using Aircrack-ng suite

Intrusion Detection & Threat Monitoring

- Deployed Snort IDS/IPS with 10+ custom signatures targeting SQL injection, XSS, and CVE-based attacks, reducing false positive rate from 35% to 8% through systematic rule tuning and threshold optimization
- Built NetFlow monitoring infrastructure on AWS EC2 using ntopng, capturing and analyzing 24+ hours of live network traffic to identify 12+ attack patterns including port scans, DDoS attempts, and data exfiltration

Security Infrastructure & Compliance

- Architected Docker-based Kerberos SSO infrastructure authenticating 2+ services (SSH, Apache) through centralized realm management, eliminating password-based authentication and reducing authentication-related security incidents by 100%
- Hardened 3+ Linux server environments implementing CIS benchmarks for SSH (key-based auth, fail2ban), Apache (mod_security, TLS 1.3), and OS-level controls, reducing attack surface by 60% as validated by OpenSCAP compliance scans
- Executed comprehensive security assessments using OpenSCAP, OSSEC, and Auditd tools identifying 40+ compliance gaps, then developed 2 NIST 800-61 incident response playbooks covering ransomware and data breach scenarios

TECHNICAL SKILLS

Security Tools: Burp Suite, Metasploit, Snort, Wireshark, Nmap, Nessus, Aircrack-ng, Hydra, GDB, pwntools, OpenSCAP, OSSEC, Auditd, GPG, OpenSSL, ntopng, IPFire, SPLUNK

Programming & Cloud: Python, Django, Bash, SQL, HTML/CSS/JavaScript, AWS EC2, GCP, Docker, Git, Kubernetes RBAC, iptables

Security Operations: Penetration Testing, Vulnerability Assessment, Binary Exploitation, OSINT, IDS/IPS, Web Application Security, Incident Response, System Hardening, SIEM Monitoring, Network Security, Log Analysis, Threat Intelligence, Security Awareness

Risk Management & Compliance: Security Risk Assessment (Quantitative & Qualitative), Risk Matrices, Risk Treatment Strategies, NIST 800-53, NIST RMF, ISO 27001, CIS Benchmarks, OWASP Top 10, Security Control Selection, Access Management, Compliance Audits, Stakeholder Communication

EDUCATION

Northeastern University, Boston, MA.

Sept 2024 – May 2026 (Expected)

Master of Science in Cybersecurity – Khoury College of Computer Sciences

Kumaraguru College of Technology, Coimbatore, India

Aug 2020 – May 2024

Bachelor of Technology in Information Technology

CERTIFICATION

CompTIA Security+: In Progress.